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Small Animal Internal Medicine Pharmacology

Treatment of Naproxen Overdose Using Therapeutic Plasma Exchange and Plasmaperfusion in a Dog

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ABSTRACT

An 8-year-old male neutered 37.4 kg mixed breed dog was presented for suspected ingestion of a large amount of naproxen (estimated at 235–294 mg/kg). The dog was treated using a combination of therapeutic plasma exchange (TPE) and plasmaperfusion (PP). The exchange of 2 plasma volumes during 140 min of TPE resulted in a 46% decrease in blood naproxen concentration. Treatment of 1 plasma volume during the following 90 min of PP resulted in an additional 10.3% decrease in naproxen concentration. The patient developed mild gastrointestinal (GI) signs that were treated medically and was successfully discharged after 4 days of hospitalization with normal follow-up laboratory test results and physical examination findings. The combination of extracorporeal treatments decreased the plasma concentration of naproxen, but the additional decrease from PP may not have provided substantial benefit compared with other conventional treatments. No reports of PP, as an alternative to hemoperfusion (HP), in veterinary medicine currently are available.

1 | Introduction

Naproxen is an over-the-counter non-steroidal anti-inflammatory drug (NSAID) that non-selectively inhibits cyclooxygenase [1]. It has oral bioavailability of 68–100%, 99% protein binding, and a low volume of distribution (Vd) of 0.12–0.13 L/kg [2, 3] In dogs, naproxen is metabolized in the liver, undergoing extensive enterohepatic circulation, and is excreted predominately in feces [2, 3]. Half-life ranges from 35 to 744 h, which is relatively long compared with other NSAIDs [1–4]. Veterinary use of naproxen is not recommended because of a narrow therapeutic index [1]. Naproxen toxicity includes GI, renal, and neurologic adverse effects [1–3]. Exact dosages for these adverse effects have not been determined, but previous literature has suggested these adverse effects occur at > 5 mg/kg, 10 to 255 mg/kg, and > 50 mg/kg, respectively [5, 6]. Additionally, naproxen has the potential to affect platelet function. Anti-platelet effects are observed in humans [7]. In dogs, altered platelet function and decreased thromboxane production have been observed with other NSAIDs [8–10]. No similar research is available for naproxen in dogs, but it is reasonable to assume ingestion of toxic amounts of naproxen could lead to platelet dysfunction.

Conventional treatment for naproxen overdose includes early induction of emesis to prevent absorption, treatments to limit absorption, and treatments to mitigate adverse effects [11–14]. The use of intralipids may be beneficial to increase the rate of excretion according to the "lipid sink" hypothesis [6, 11, 12]. With increasing availability, extracorporeal blood purification, including therapeutic plasma exchange (TPE) and hemoperfusion (HP), is also a potential treatment for NSAID toxicity [5, 11, 15–22]. Plasmaperfusion (PP) combines these two

Abbreviations: AC, activated charcoal; ACT, activated clotting time; BUN, blood urea nitrogen; HP, hemoperfusion; NSAID, non-steroidal anti-inflammatory drug; PP, plasmaperfusion; PV, plasma volume; RI, reference interval; SDMA, symmetric dimethylarginine; TPE, therapeutic plasma exchange; Vd, volume of distribution.

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modalities wherein patient plasma is separated, exposed to a carbon filter, and then returned to the patient. This process allows for additional drug removal while conserving the patient's plasma. To our knowledge, the combination of TPE with PP has not been previously reported in a dog.

2 | Case Presentation

An 8-year-old male neutered 37.4 kg mixed breed dog was presented to a specialty private practice and emergency center for reported consumption of 235 to 2944 mg/kg of naproxen within the prior 13 h. The dog had no relevant historical medical abnormalities. The owner reported no clinical signs before presentation. Physical examination was unremarkable. Emesis was induced with apomorphine (0.01 mg/kg IV; Stokes Pharmacy, Mt. Laurel Township, NJ) which produced vomiting of a moderate amount of digested food, but no identifiable naproxen tablets. A single dose of activated charcoal without sorbitol (10 mL/kg PO; Toxiban, Vet-A-Mix, Lloyd, Sioux Falls, SD) was administered. An IV catheter was placed for administration of maropitant (1 mg/kg IV; Cerenia, Zoetis, Parisppany-Troy Hills, NJ) and lactated Ringer's solution (3.6 mL/kg/h IV; Hospira, Pfizer, Lake Forest, IL).

Initial laboratory testing indicated a packed cell volume of 55% and total protein concentration of 7.0 g/dL. Blood gas analysis (EPOC Blood Analysis System, Siemens Healthineers. Erlanger, Germany) was normal, with blood urea nitrogen (BUN) concentration of 18 mg/dL (reference interval [RI], 7–26 mg/dL) and serum creatinine concentration of 0.87 mg/dL (RI, 0.4–1.5 mg/dL). Results of a CBC (ProCyte Dx Veterinary CBC Hematology Analyzer, Idexx Laboratories. Westbrook, ME) and serum biochemistry (Catalyst One, Idexx Laboratories, Westbrook, ME) were unremarkable (symmetric dimethylarginine [SDMA], 12 μg/dL; RI, 0–14 μg/dL; BUN, 19 mg/dL; RI, 9–31 mg/dL; creatinine, 1.2 mg/dL; RI, 0.5–1.5 mg/dL). Urine specific gravity was > 1.050.

The dog was sedated using butorphanol (0.4 mg/kg IV; Dolorex, Merck Sharpe & Dohme, Rahway, NJ) and dexmedetomidine (2.5 mcg/kg IV; Dexmedesed, Dechra, Norwich, United Kingdom). A 14 Fr × 30 cm double lumen silicone dialysis catheter (Hemo-Cath ST, Medical Components Inc., Harleysville, PA) was placed in the right jugular vein using the transcutaneous modified Selinger technique. The TPE was performed using a PRISMA TPE 2000 and Thermaxx warming circuit on a Prismax machine (Baxter International Inc. Deerfield, IL). The total circuit volume approximated 4.5% of the patient's blood volume based on an estimated blood volume of 3366 mL and a circuit volume of 152 mL. The plasma volume (PV) was calculated as 1.35 L based on the equation ([100-Hct] × [0.8 × body weight in kilograms]). A total exchange of 2 plasma volumes was performed over 140 min with the intention to decrease the naproxen concentration by approximately 86% [23]. The 2.6 L of removed PV was replaced with 1600 mL of fresh frozen plasma (FFP; in-hospital blood bank), 200 mL of lactated Ringer's solution, 300 mL of 0.9% saline (Baxter International Inc. Deerfield, IL) and 500 mL of 6% hydroxyethyl starch (Vetstarch, Zoetis, Parisppany-Troy Hills NJ).

At minute 140, a pre-primed carbon adsorbent was added to the circuit (Dublin 300 mL, Aimalojic. Rapid City, SD). The adsorbent inlet side was attached to the TPE circuit effluent line, which contained separated patient plasma. The adsorbent outlet side was attached to a sterile bag that collected post-adsorbent plasma, which was returned to the patient. Plasmaperfusion was performed for 90 min to treat one additional PV. Blood flow rate was 170 to 2200 mL/min and the post-filter replacement fluid rate was 1100 mL/h, resulting in a filtration fraction of 18% to 23%. Systemic heparinization was achieved using unfractionated heparin (Techdow USA Inc., Langhorne, PA) via an initial bolus of 50 U/kg followed by a constant rate infusion of 25 U/kg/h. The heparin rate was adjusted based on activated coagulation time (ACT) monitoring targeted at 180 to 2200 s (Medtronic ACT Plus Automated Coagulation Timer, MAG Medical Equipment, Boise, ID). Calcium gluconate (calcium gluconate 23%, VetOne, Boise, ID), diluted in 0.9% saline to 46 mg/mL, was administered via a peripheral IV catheter for the duration of TPE and then discontinued during PP. The rate of calcium replacement was adjusted to maintain a serum ionized calcium concentration of 0.9 to 1.22 mmol/L (RI, 1.1–1.4 mmol/L). The dog remained stable cardiovascular function throughout treatment. Two doses of dexmedetomidine (2.5 mcg/kg IV) were given to facilitate cooperation. Shortly after the first unit of FFP, mild facial angioedema was noted and improved after administration of diphenhydramine (2 mg/kg SC; Armas Pharmaceuticals, Freehold, NJ).

Samples were collected from the TPE access, TPE effluent, and post-PP lines (when available) 30 min (after induction of TPE), 140 min (end of TPE), 150 min (start of PP), 240 min (end of PP) and 12 h after treatment (Table 1). Samples were stored at −18°C for 2 months before submission. Results were reported for unbound naproxen and partial protein-bound concentration based on liquid chromatography—mass spectrometry (NSAID Drug Screen, Texas A&M Veterinary Medical Diagnostics Lab, College Station, Tx).

After treatment, the dog was maintained on IV lactated Ringer's solution (2 mL/kg/h IV), misoprostol (2.3 mcg/kg PO q8h; Greenstone LLC, Gladstone, NJ), omeprazole (0.9 mg/kg PO q12h; Glenmark Pharmaceuticals, Mumbai, India), and trazodone (5 mg/kg PO as needed; Teva Pharmaceuticals, Tel Aviv-Yafo, Israel). Laboratory testing was performed daily. Kidney and liver function remained normal. A large volume of melenic diarrhea was passed the first night and progressed to moderate volume light brown diarrhea with mild hematochezia. The patient had one episode of vomiting and intermittent hyporexia during hospitalization. Additional treatment included probiotics (Visbisome 2 capsules PO q12h; ExeGI Pharma, Rockville, MD), ondansetron (0.5 mg/kg IV q12h; Fosun Pharma, Shanghai, China) and metronidazole (11.3 mg/kg PO q12h; Unichem Laboratories, Mumbai, India). The patient was discharged on day 4 of hospitalization with improved clinical signs. Reevaluation was performed 5 days later by the primary care veterinarian, and laboratory test results remained normal.

3 | Discussion

Extracorporeal blood purification methods have been used to treat acute NSAID ingestion in dogs [5, 11, 15–22]. Therapeutic plasma exchange is performed by separating and removing a

TABLE 1 | Naproxen drug concentration.

Time	Treatment	Patient (μg/ml)	Effluent (μg/ml)	Post PP plasma (μg/ml)	% Serum reduction per unit time	% Serum reduction overall
30 min	TPE	27.2	—	—	—	—
140 min	TPE	14.5	—	—	46	46.7
150 min	PP	17.5	10.7	6.2	42	35.7
230 min	PP	15.7	11.9	7.2	39	42.3
12 h post treatment	—	15.4	—	—	—	43.4

Note: Naproxen drug concentration in patient effluent and post-plasmaperfusion plasma. The first 140 min were performed under TPE with a 10-min transition time. The final 90 min were performed under plasmaperfusion. Finally, drug concentrations were collected 12 h post-treatment.

patient's plasma from the remaining blood components, which are returned to the patient. This process is effective at removing unwanted proteins, immune complexes, autoantibodies, and protein-bound substances, such as toxins [5, 11, 16–19, 21, 22, 24–27]. Highly protein-bound substances (> 80%) with small volume of distribution (< 0.2 L/kg), such as naproxen and other NSAIDs, can be effectively removed [5, 11, 16–19, 21, 22, 25, 28]. The volume of plasma exchanged during TPE is based on the patient's plasma volume (PV) and the desired amount of substance removal. Using first-order kinetics, it is generally expected that a substance should decrease by approximately 63%, 78%, and 86% after 1.0, 1.5, and 2.0 PV exchanges, respectively [23]. Plasma volume exchanges > 2.0 result in smaller and less clinically relevant decreases while increasing cost and transfusion exposure [23, 24, 28].

Therapeutic plasma exchange is not without disadvantages. Beyond the cost of equipment and the need for advanced training, removing plasma requires replacement fluids. Colloid replacement using 3%–5% albumin generally is preferred in humans because FFP is associated with increased complications [23, 24, 29, 30]. In veterinary medicine, the use of canine albumin is limited by availability and cost. Human albumin can be used in dogs but may have substantial adverse effects [31, 32]. Alternatives for plasma replacement are typically limited to synthetic colloids or FFP. The risk of acute kidney injury and increased mortality with synthetic colloids is well documented in humans [33, 34]. In veterinary medicine, some studies have identified negative effects whereas others have not [35–38]. Additionally, synthetic colloids have potential effects on coagulation, but their extent and clinical relevance remain controversial [39–43]. In veterinary medicine, FFP also may be limited because of cost, availability, and the complications noted above [26, 29, 30]. Although a retrospective evaluation of TPE found that FFP was associated with increased complications, all but one were mild and self-limiting [26]. A similarly mild reaction noted in our patient resolved with diphenhydramine treatment. In our case, the replacement fluid initially was planned as 55% FFP, 31% crystalloid, and 19% synthetic colloid. Because of unanticipated larger units, FFP replacement was 61%. Larger volume FFP replacement still was selected based on the potential delay in presentation from ingestion and concern for the development of gastrointestinal signs that could lead to further protein loss after treatment. Although a portion of replaced synthetic colloids is removed with TPE, their use is purposefully limited to minimize additional risk of renal injury or hypocoagulability.

Hemoperfusion targets the elimination of toxic substances without the need for replacement fluids. Whole blood is exposed to an adsorbent to bind the substance of interest, including NSAIDs [15, 16, 20, 44]. Although HP is effective for protein-bound substances, adsorbent saturation decreases the efficacy of treatment over time [45]. Adsorbent cartridges often are replaced during treatment to avoid ineffective substance removal, but doing so adds cost. Additionally, timing for replacement is mostly unknown. Plasmaperfusion, also referred to as plasma adsorption or plasma filtration-adsorption, is a variation of HP in which only plasma is exposed to the adsorbent. This modality provides improved hemocompatibility without the requirement of replacement fluids and has been performed in humans for the removal of cytokines, inflammatory mediators, and toxins [45, 46]. Because of suspected marked exposure in our case, TPE initially was chosen, initially HP to avoid potential sorbent saturation. However, there was concern that, even with 86% reduction, sub-toxic concentrations would not be achieved. Given that TPE already provided a method of plasma separation, PP was added for an additional 1 PV.

The first naproxen concentration obtained was 27.2 μg/mL. Unfortunately, this sample was taken 30 min into TPE, and no baseline sample was obtained. A previous study reported naproxen concentrations of 48.9 μg/mL at initiation and 42.8 μg/mL 30 min into treatment for a 45 mg/kg ingestion [18]. The previously reported blood naproxen concentrations were much higher compared with those in our patient despite substantially lower reported ingestion. In both cases, approximately 45% of 1 PV would have been exchanged in 30 min. Although two patients cannot be directly compared, we suspect our baseline concentration would not have been above 48.9 μg/mL. Potential reasons for the lower-than-expected concentration are either that our patient ingested less naproxen than reported or previous decontamination was highly effective. In the absence of real-time drug concentrations, treatment decisions must be made based on the maximum possible ingested dose.

Therapeutic plasma exchange resulted in a 46% decrease in naproxen concentration. This result is much lower than the expected 86% with 2 PV exchanges [23, 28]. The calculated

reduction may have been slightly higher had a baseline naproxen concentration been available. Expected changes in drug concentrations during TPE are based on one-compartment models in which no movement into or out of the intravascular space occurs [23]. Although the pharmacokinetics of naproxen make redistribution out of the intravascular space unlikely, overdoses may result in saturated protein binding and endogenous clearance pathways that alter these kinetics [2–4, 28]. The surface area of TPE filters is determined by the manufacturer, but it can decrease during treatment if clotting, pore obstruction or both occur [25]. The absence of visible clotting and stable transmembrane pressure make the presence of micro-clots or pore obstruction unlikely in our case. Because of the large pore size and sieving coefficient, plasma movement is not limited in modern TPE filters and likely was not a factor [24]. Because the 74-h half-life of naproxen markedly exceeds our 3 h and 50 min duration of treatment, it is unlikely that any endogenous clearance would have contributed to the lower-than-expected concentrations.

A 20% increase in serum naproxen concentration was noted from minutes 140 to 150. This increase could have represented redistribution into the intravascular space during the transition from TPE to PP, although this explanation would be inconsistent with the one-compartment model. The short time frame of only 10 min made additional gastrointestinal absorption or enterohepatic recirculation unlikely. Intra-assay variability also could have occurred. Because of laboratory methodology, the extent of variability of the assay was not reported. Plasma perfusion of an additional 1 PV decreased the plasma naproxen concentration by another 10.3% over 90 min. The efficacy of PP was expected to be lower because of the smaller PV treated and lower starting plasma naproxen concentration. If PP had been performed earlier, or throughout the session, a larger decrease may have been seen. Measured effluent concentrations of naproxen at 150 and 240 min were decreased compared with patient serum concentration before carbon adsorption. These two results would be expected to be similar, but adsorption to the TPE filter could have occurred or a fraction of the toxin could have been bound to other non-plasma components.

The 12-h post-treatment patient plasma naproxen concentration (15.4 μg/mL) was similar to the concentration at the end of treatment (15.7 μg/mL), which suggests either prolonged half-life because of enterohepatic recirculation or inability of the assay to detect small changes in plasma concentration. Only one dose of activated charcoal (AC) was administered in this case because of the risk of aspiration and hypernatremia with repeated AC adminisration [47]. Routine use of AC is not recommended in human medicine given a lack of evidence showing improved morbidity and mortality [48, 49]. Although the half-life is shorter, single dose administration of AC is as effective as multidose administration for treatment of carprofen ingestion [50]. Given the drug concentrations in the dog of our report, additional AC may have been beneficial in shortening drug half-life. The use of metronidazole in our patient also could be questioned because its use in dogs with acute diarrhea lacks support [51]. Intestinal bacteria, particularly gram-negative bacteria, may contribute to NSAID-induced enteropathy, and antibiotics show a protective effect in rat models [52–54]. Metronidazole has protective benefits in patients with NSAID enteropathy, including decreased inflammation and intestinal barrier permeability [53–57]. Whether the use of metronidazole in our patient was beneficial is unknown.

Limitations of our case report include lack of baseline samples and the limited number of samples tested. More frequent samples would have been helpful for documenting the kinetics of drug removal and intra-assay variation, but were cost prohibitive. The months-long delay in sample submission and storage in a non-temperature-monitored freezer makes degradation as a possibility. Naproxen is relatively stable with a shelf life beyond its expiration date, but rates of degradation in frozen plasma samples are not established [58].

Our report shows that the combination of TPE and PP is feasible and may be used to decrease the plasma concentration of naproxen and potentially other NSAIDs in acute toxicity. Given the lower percentage of drug removal with PP, it remains unclear if its addition had a clinically relevant benefit or if instituting PP earlier would have led to similar total drug removal with fewer TPE PV exchanges. Future studies are warranted to further investigate the role of PP in veterinary medicine.

Disclosure

The antimicrobial metronidazole was used in the dog. It might have benefit associated with non-steroidal anti-inflammatory drug (NSAID) associated enteropathy but its use for diarrhea is not supported by literature.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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